

Roll No: | | | | | | | | | | | |

B. TECH — III SEMESTER • NOVEMBER, 2023

Time: 1 Hour

Max. Marks: 30

Q. No.	Question	Max. Marks	CO(s)
1. (a)	What is a group and semigroup?	2	CO3
(b)	What is Euler circuit and Hamiltonian circuit?	2	CO4
(c)	Explain isomorphism with an example.	2	CO3
(d)	What is homomorphism?	2	CO3
(e)	What is graph coloring and chromatic number of a graph?	2	CO4
2. (a)	Explain cosets and Lagrange's theorem.	5	CO3
(b)	Explain groups and coding in detail.	5	CO3
3. (a)	Explain Breadth-First Search (BFS) with a suitable example.	5	CO4
(b)	Explain Depth-First Search (DFS) with a suitable example.	5	CO4
4. (a)	Define and explain Minimum Spanning Tree (MST) with an example.	5	CO4
(b)	State and explain the Five Color Theorem for planar graphs.	5	CO4